AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS 2021

Physics

Total Time: 2 hours 10 minutes

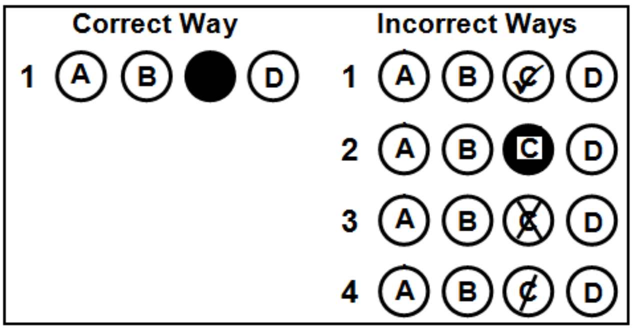
Total Marks: 65 (50-Theory & 15-Alternate to Practical)

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

<u>THEORY</u> (Questions 1-50)

1. The dimensions of momentum are represented as

   A. $[ML^2T^{-1}]$
   B. $[MLT^{-1}]$
   C. $[MLT^{-2}]$
   D. $[ML^{-1}T]$

2. Uncertainty in measurement can occur due to the

   I. limitation of instruments
   II. natural variations
   III. frequent errors

   A. I only.
   B. III only.
   C. I and II.
   D. II and III.

3. Random errors can be reduced by

   I. taking zero correction
   II. taking mean of several measurements
   III. comparing the instrument with another more accurate one

   A. II only.
   B. III only.
   C. I and II.
   D. I and III.

4. The CORRECT scientific notation for 100.00 is

   A. 1.00×10^{-3}
   B. 1.00×10^{-2}
   C. 1.00×10^{2}
   D. 1.00×10^{3}

5. If the dot product of two vectors (**X** and **Y**) is zero, then which of the following conditions is TRUE?

   A. Both vectors are opposite in direction
   B. Both vectors are parallel to each other
   C. Either of the two vectors is a unit vector
   D. Either of the two vectors is a null vector

6. The cross product of two vectors (**X** and **Y**) is a negative vector when the angle between them is

   A. 0^o
   B. 90^o
   C. 180^o
   D. 270^o

7. Torque has minimum value when the angle between force **F** and force arm **r** is

   A. 0^o
   B. 30^o
   C. 60^o
   D. 90^o

8. In a Cartesian plane, the *x*-axis is called

   A. *z*-axis.
   B. vertical axis.
   C. rotational axis.
   D. horizontal axis.

9. If the value of moment arm is zero, then the value of torque will be

   A. $1 > \tau > 0$
   B. $\tau < 0$
   C. $\tau = 0$
   D. $\tau = 1$

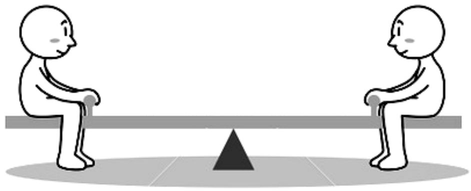
10. If the magnitude of a vector **Y** is the same as that of vector **X** but opposite in direction, then vector **Y** is equal to

   A. 1
   B. 0
   C. **X**
   D. – **X**

11. According to the head to tail rule, if the forces are unequal in terms of direction and magnitude, then the minimum number of forces with the sum of zero will be

   A. 2
   B. 3
   C. 4
   D. 5

12. The given figure depicts one of the conditions of equilibrium.



Which of the following options identifies the CORRECT name and statement of the condition?

	Name of Condition	Statement
A	First	Vector sum of all the torques is zero
B	Second	Vector sum of all the forces is zero
C	First	Vector sum of all the momenta is zero
D	Second	Vector sum of all the torques is zero

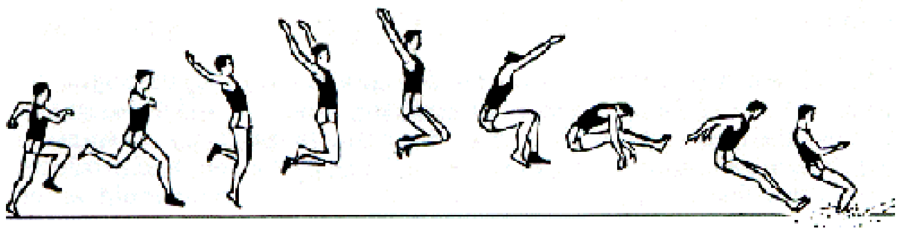
13. If the slope of a velocity-time graph gradually increases, then the body is said to be moving with

A. uniform velocity.
B. positive acceleration.
C. negative acceleration.
D. instantaneous velocity.

14. A lighter metallic ball (m_1) collides with the much heavier metallic ball (m_2) at rest, when ($m_1 << m_2$), then after collision, the heavier ball (m_2) will

A. remain at rest.
B. gain the lighter ball's velocity.
C. move in the opposite direction.
D. move with double the velocity of lighter ball.

15. The given figure shows the first attempt of a long jump of an athlete in Olympics.



Which of the following factors will MAINLY help him to break his own record in the next attempt?

A. Angle of projection of the athlete
B. Acceleration of the athlete
C. Body shape of the athlete
D. Weight of the athlete

16. Missiles **I**, **II**, **III** and **IV** are launched at 35º, 45º, 55º and 65º, respectively from the origin.

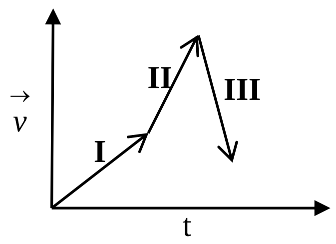
The shortest distance will be covered by

A. Missile I.
B. Missile II.
C. Missile III.
D. Missile IV.

17. A two dimensional motion under the acceleration due to gravity is known as

A. linear motion.
B. circular motion.
C. vibratory motion.
D. projectile motion.

18. In the given velocity-time graph, the body is said to have



A. zero acceleration in region I.
B. variable acceleration in region II.
C. positive acceleration in region III.
D. negative acceleration in region III.

19. If a water droplet falls from a height in air, its terminal velocity will be attained when the drag force is

A. twice the weight of droplet.
B. thrice the weight of droplet.
C. half of the weight of droplet.
D. equal to the weight of droplet.

20. All of the following are applications of Bernoulli's effect EXCEPT

A. aerofoil.
B. atomizer.
C. venturi meter.
D. vacuum cleaner.

21. If each particle of a fluid passing through a point follows the same path, then this flow is called

    A. ideal flow.
    B. normal flow.
    C. turbulent flow.
    D. streamline flow.

22. According to the equation of continuity, if the cross-sectional area of a pipe decreases, then the speed of the fluid passing through it

    A. increases.
    B. decreases.
    C. remains the same.
    D. varies unpredictably.

23. The CORRECT statement in the light of Bernoulli's theorem is that the

    A. speed of the fluid is low at low pressure.
    B. speed of the fluid is high at low pressure.
    C. pressure of a fluid is independent of its speed.
    D. theorem is valid only for turbulent flow of a liquid.

24. When an aeroplane is flying, the air moves from the top and the bottom of the wings. The air moving over the top of the wings is faster due to which the pressure above the wings will

    A. be low.
    B. be high.
    C. be zero.
    D. vary unpredictably.

25. If a simple pendulum is shifted from Karachi to the peak of K-2 mountain, then its time period

    A. increases.
    B. decreases.
    C. becomes zero.
    D. remains the same.

26. Loud music produced by beating wooden drum is an example of

    A. beats.
    B. free vibrations.
    C. forced vibrations.
    D. damped oscillations.

27. If the amplitude of a simple pendulum is increased four times, then its time period will

    A. reduce to half.
    B. remain the same.
    C. increase two times.
    D. decrease four times.

28. When a car moves on any uneven surface, then its shock absorbers play an important role.

    The working of shock absorbers exemplifies

    A. free oscillation.
    B. force oscillation.
    C. damped oscillation.
    D. undamped oscillation.

29. When the bob of a simple pendulum is at the mean position during oscillation, its kinetic energy will

    A. be zero.
    B. be minimum.
    C. be maximum.
    D. remain constant.

30. All of the following are the examples of free oscillation EXCEPT

    A. prongs of a tuning fork.
    B. bob of a simple pendulum.
    C. motion of a child on a swing.
    D. vibrating wire of a sonometer.

31. Which of the following quantities of sound is affected by the change in air temperature?

    A. Intensity
    B. Loudness
    C. Amplitude
    D. Velocity

32. Which of the following statements about the speed of waves on a string is/ are TRUE?

    I. The speed depends on the frequency.
    II. The speed depends on the tension in the string.
    III. The speed depends on the mass per unit length of the string.

    A. I only
    B. II only
    C. I and III
    D. II and III

33. If a wave has a frequency of 10 Hz, then the time period of the wave will be

    A. 0.1 s.
    B. 1 s.
    C. 10 s.
    D. 100 s.

34. When two tuning forks of nearly the same frequencies are sounded together, a note of alternately increasing and decreasing intensity will be heard.

This note is called

A. beat.
B. waveform.
C. diffraction.
D. polarisation.

35. The option that shows the conditions used by Laplace for correcting the velocity of sound is

	Medium Used	**Thermodynamics Process**
A	Solid	Adiabatic
B	Liquid	Isobaric
C	Gas	Adiabatic
D	Gas	Isothermal

36. If two sound waves with frequencies of 128 Hz and 126 Hz are produced simultaneously by two different tunning forks, then the detected beat will have a frequency

A. 2 Hz.
B. 126 Hz.
C. 128 Hz.
D. 256 Hz.

37. When a listener moves away from a stationary source of sound, then for the listener the apparent change in the pitch and frequency of sound is

	Pitch	**Frequency**
A	low	low
B	high	low
C	low	high
D	high	high

38. When a swimmer goes deeper into the swimming pool, the temperature of water decreases. As a result, the speed of sound will

A. increase.
B. decrease.
C. become zero.
D. remain the same.

39. An application of Doppler's effect that is used to measure the depth of ocean is

    A. radar.
    B. sonar.
    C. red shift.
    D. blue shift.

40. When an ideal gas is cooled at a constant volume till the pressure reduces to half of the original pressure, the average translational kinetic energy of the gas molecules will

    A. remain unchanged.
    B. become half of the original.
    C. become two times of the original.
    D. become three times of the original.

41. All Carnot engines have zero efficiency if they are operating between two bodies with

    A. the same temperature and the same nature of working substance.
    B. different temperatures and the same nature of working substance.
    C. the same temperature irrespective of the nature of working substance.
    D. different temperatures irrespective of the nature of working substance.

42. Entropy can be described as the

    A. decrease in energy in the system.
    B. increase in energy in the surrounding.
    C. unavailability of energy in the system.
    D. degradation of energy in the environment.

43. Which of the following relations describes the Boyle's Law?

    A. Volume is directly proportional to temperature
    B. Pressure is directly proportional to temperature
    C. Pressure is inversely proportional to volume at constant temperature
    D. Number of moles is inversely proportional to volume at constant temperature

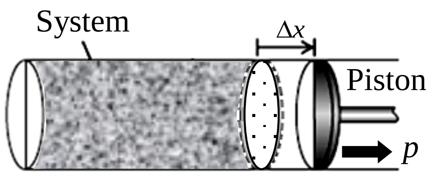
44. If the volume of a thermodynamics system increases, then the work done by the system will be

    A. zero.
    B. infinite.
    C. positive.
    D. negative.

45. In a thermodynamic system, if there is no change in temperature and internal energy, then the process is called an

    A. isobaric process.
    B. adiabatic process.
    C. isochoric process.
    D. isothermal process.

46. The given figure shows a cylinder containing fluid with a movable piston.



If the piston moves out a small distance Δx, then the work done by the gas is

A. $-p\Delta x$
B. $-pA\Delta x$
C. $p\Delta x$
D. $pA\Delta x$

47. The efficiency of Carnot Cycle depends upon the change in the

A. mass of active substance.
B. volume of active substance.
C. pressure on the hot reservoir.
D. temperature of hot and cold reservoirs.

48. Water freezing into ice is an example of

A. zero entropy.
B. positive entropy.
C. variable entropy.
D. negative entropy.

49. When any two objects are rubbed together in thermodynamics, their internal energy

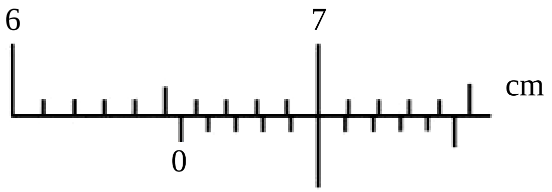
A. increases.
B. decreases.
C. becomes zero.
D. remains constant.

50. All of the following are the examples of reversible process EXCEPT

A. boiling of raw eggs.
B. compression of a helical spring.
C. slow compression of oxygen gas.
D. freezing of a carbonated drink into ice.

<u>ALTERNATE TO PRACTICAL</u> (ATP: Questions 51-65)

51. The given diagram shows two different scales on a Vernier callipers.



The reading on the Vernier scale is

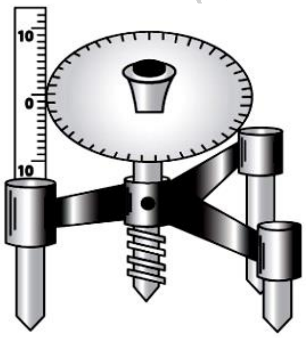
A. 0.5
B. 5
C. 6.5
D. 7

52. A bag contains 50 metallic spheres of different diameters. A task is given to students to classify these spheres in terms of their diameters and find the number of spheres that lie between the range of 0.3 cm and 1.5 cm out of 50 metallic spheres.

For classifying as accurately as possible, the MOST appropriate instrument the students should pick from the science lab is

A. spherometre.
B. screw gauge.
C. measuring tape.
D. Vernier callipers.

53. The given image shows a measuring instrument.

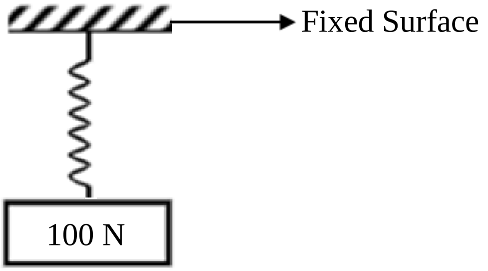


On the main scale of this instrument, the purpose of markings above and below the zero mark is to measure the

I. depth.
II. height.
III. thickness.

A. I only
B. III only
C. I and II
D. II and III

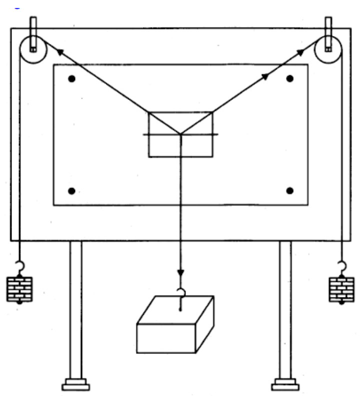
54. The given diagram shows a helical spring attached to a fixed surface. The value of spring constant is taken as ‘k’ 300 N/m and a 100 N weight is attached at the bottom.



The extension in the spring will be

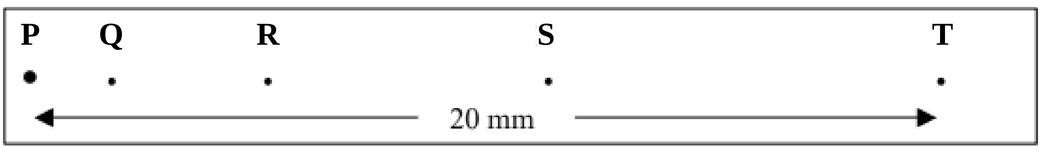
A. 0.333 m.
B. 0.666 m.
C. 3 m.
D. 6 m.

55. Which of the given sources of errors will NOT affect your readings while handling the Gravesand’s apparatus?



A. Friction in the pulleys
B. Use of rusty holding pins
C. Use of inaccurate weights
D. Placement of the drawing board

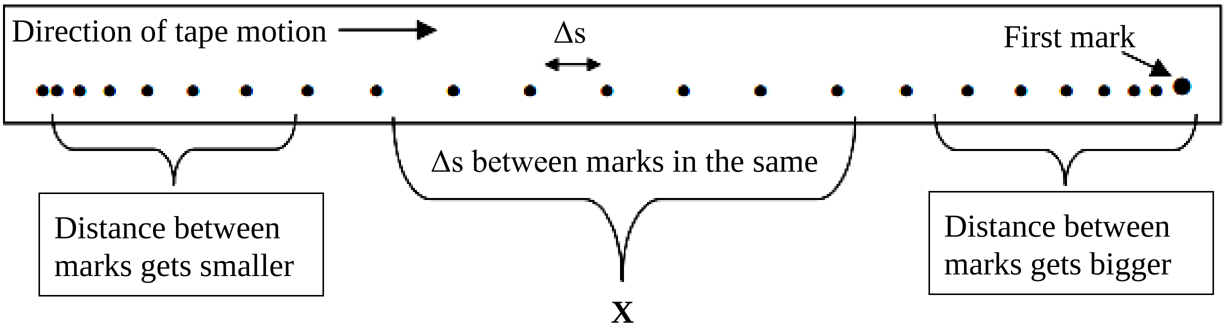
56. The given diagram shows a ticker tape. The frequency of the ticker tape is 20 Hz.



How much time will be taken by the ticker tape to plot dots from **P** to **T**?

A. 0.05 s.
B. 0.25 s.
C. 5 s.
D. 20 s.

57. The given figure shows dots on a ticker tape after performing an experiment.



The value of acceleration for the bracket region **X** will be

A. $a < 0$
B. $a = 0$
C. $a > 0$
D. $0 < a < 1$

58. If a student plans to make a simple pendulum of time period 1 s, then the length of the pendulum will be

(**Note**: Take the value of acceleration due to gravity as 10 m/s^2.)

A. 0.25 m.
B. 0.50 m.
C. 1.00 m.
D. 2.00 m.

59. A second pendulum is oscillating in an elevator.

If the elevator falls freely towards the Earth, then the time period of the pendulum will become

A. 0 s.
B. 1 s.
C. 2 s.
D. ∞.

60. A student performs an experiment in his college laboratory to determine the specific heat of water by electrical method. At the end of experiment, the graph that he drew between temperature (T) and time (t) was incorrect.

Which of the following can be a source of error that caused inaccurate reading during his observation?

A. Loss of heat due to radiation
B. Large increase in temperature of water
C. Greater least count of the thermometer
D. Non-steady voltage of the power supply

61. A cylindrical resonance tube, opened at both ends, has a fundamental frequency ‘f’ in air as shown in the given **figure I**.

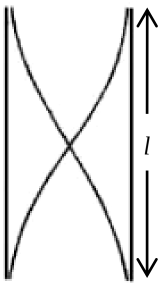


Figure I

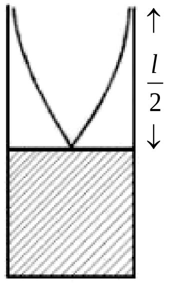


Figure II

As shown in **figure II**, if half of the length is dipped vertically into water, then the fundamental frequency of the air column will become

A. $\frac{f}{2}$
B. f
C. $\frac{3f}{2}$
D. $2f$

62. The distance between two consecutive nodes of an incident wave is 2 m.

If the frequency of the wave is 50 Hz, then the speed of this wave is

A. 25 m/s.
B. 50 m/s.
C. 100 m/s.
D. 200 m/s.

63. The given observation table is used to determine the value of mechanical equivalent of heat.

Time (t)	**Potential Difference (V)**	**Electric Current (I)**	**Temperature (T)**
2 seconds	5 volts	2 amperes	2^oC

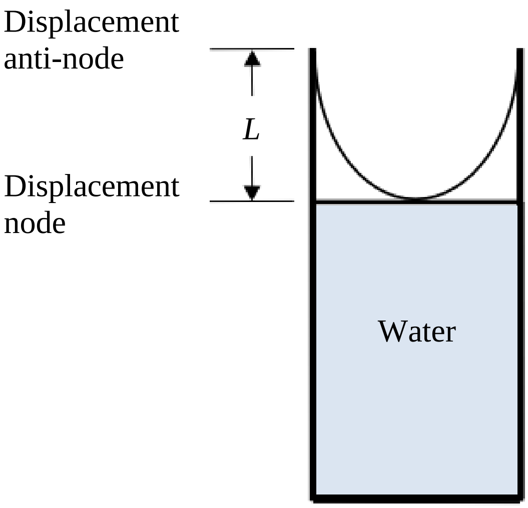
The electrical energy (W) consumed will be

A. 5 J.
B. 10 J.
C. 20 J.
D. 40 J.

64. All of the following are precautions which should be taken into consideration while measuring the mechanical equivalent of heat by electric method EXCEPT that the

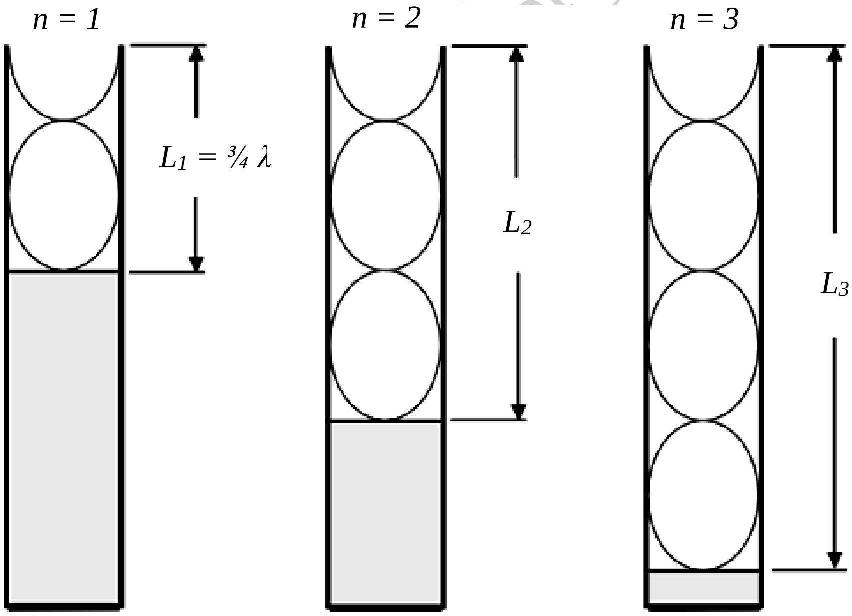
A. supply of voltage is steady.
B. heating coil should be immersed in water.
C. calorimeter should be placed in an insulation box.
D. thermometer should touch the base of the calorimeter.

65. Suppose that a sound wave is travelling through a given resonance tube.



At a constant temperature, the speed of sound and the frequency of a tuning fork remain constant and hence, the wavelength of the sound wave should also be constant.

In the given condition, when the length of the tube becomes L_2 and L_3, the length of the air column will be



	L_2	**L_3**
A	5/4 λ	7/4 λ
B	3/4 λ	3/4 λ
C	3/4 λ	7/4 λ
D	7/4 λ	7/4 λ

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